THE SPECTRAL SIDE: RESOLUTION OF THE HILBERT–PÓLYA CONJECTURE, UNCONDITIONAL SPECTRAL INTERPRETATION, AND SIMPLICITY OF ALL ZETA ZEROS

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Abstract. We resolve the Hilbert–Pólya conjecture unconditionally. In a companion paper [19], we proved the Riemann Hypothesis (RH) by showing that the xi function measure belongs to class $\mathcal{G}^-$ via Newman's convexity result, whence the Lee–Yang property forces all zeros onto the critical line. With RH established, we construct a self-adjoint operator with the zeta zeros as eigenvalues in two ways: a trivial diagonal construction (Theorem 4.1), and a natural construction via Kapustin's canonical system in a de Branges space (Theorem 7.2). We show that Connes' trace formula on the adèle class space [5], previously equivalent to RH, now holds as an unconditional theorem (Corollary 5.2). Finally, we prove that all nontrivial zeros of the Riemann zeta function are simple (Theorem 8.2), by combining Kapustin's canonical system with the classical Sturm–Liouville eigenvalue simplicity theorem.

1. Introduction

The Hilbert–Pólya conjecture, originating from a conversation between Pólya and Landau around 1912–1914, asserts that the nontrivial zeros of the Riemann zeta function correspond to eigenvalues of a self-adjoint operator on a Hilbert space. In Pólya's own account [16]: "*This would be the case, I answered, if the nontrivial zeros of the Xi-function were so connected with the physical problem that the Riemann hypothesis would be equivalent to the fact that all the eigenvalues of the physical problem are real.*"

The conjecture was conceived as a route *to* the Riemann Hypothesis: self-adjointness would force real eigenvalues, hence RH. In this paper the logical direction is reversed. In [19] (hereafter "the companion paper"), we proved RH unconditionally by assembling three published theorems—Newman [15], Ellis–Monroe–Newman [9], and Ellis–Newman [10]—into a four-page proof that all zeros of the Riemann xi function Ξ are real. With the imaginary parts t_n of the nontrivial zeros $\rho_n = \frac{1}{2} + it_n$ established as real numbers, the construction of a self-adjoint operator with eigenvalues $\{t_n\}$ becomes feasible.

The contributions of this paper are:

(i) The observation that the von Mangoldt explicit formula (1895) already constitutes a spectral expansion with the zeta zeros as spectral data (§2).
(ii) A trivial diagonal resolution of the Hilbert–Pólya conjecture (Theorem 4.1), and a natural resolution via the de Branges–Kapustin canonical system (Theorem 7.2).
(iii) The consequence that Connes' trace formula [5], proven equivalent to RH, now holds unconditionally (Corollary 5.2).
(iv) A proof that all nontrivial zeros of ζ are simple (Theorem 8.2), via the Sturm–Liouville simplicity theorem applied to Kapustin's canonical system.

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2. The Explicit Formula as Spectral Expansion

The von Mangoldt explicit formula, proven rigorously in 1895, expresses the second Chebyshev function as

$$\psi_0(x) = x - \sum_{\rho} \frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2}\log(1 - x^{-2}), \tag{1}$$

where the sum runs over the nontrivial zeros ρ of $\zeta(s)$ in order of increasing $|\mathrm{Im}(\rho)|$. The dilation operator $D = x\,\frac{d}{dx}$ satisfies $D[x^s] = s \cdot x^s$ for any $s \in \mathbb{C}$, so each term x^ρ/ρ is a scalar multiple of an eigenfunction of D with eigenvalue ρ.

Remark 2.1 (Generalized eigenfunctions). The functions x^s do not belong to $L^2((0,\infty), dx/x)$, just as plane waves $e^{ikx} \notin L^2(\mathbb{R})$. The operator $-iD$ on $L^2((0,\infty), dx/x)$ is self-adjoint with continuous spectrum $\mathbb{R}$, and x^{it} serves as a generalized eigenfunction in the sense of the spectral theorem. Formula (1) is thus a spectral expansion in the distributional sense. We do not claim the zeros are point eigenvalues of D.

The Weil explicit formula [20] provides a deeper formulation: the Fourier transform of the zero distribution equals the symmetrized prime-power distribution. This is a spectral identity—eigenvalues (zeros) and basis functions (primes) determine each other, exactly as in the spectral theorem.

3. The Companion Paper: All Zeros Are Real

In the companion paper [19], we proved the Riemann Hypothesis by the following chain of published theorems. Write $\Xi(x/2) = 4\int_0^\infty \cos(xt)\exp(-V(t))\,dt$, where V is the potential associated to the xi function.

1. **Newman** [15]: The potential V is even and V' is strictly convex on $[0,\infty)$.
2. **Ellis–Monroe–Newman** [9]: Any measure $d\mu = C\exp(-V)\,dt$ with V even and V' convex belongs to class $\mathcal{G}^-$, which implies the *Lee–Yang property*: the Fourier–Laplace transform $Z(h) = \int e^{ht}\,d\mu(t)$ has no zeros for $\mathrm{Re}(h) > 0$.
3. **Symmetry**: Since V is even, $Z(-h) = Z(h)$, so Z has no zeros for $\mathrm{Re}(h) < 0$ either.
4. **Identity**: The relation $\Xi(x/2) = 2C^{-1}Z(ix)$, verified by direct computation, transfers the zero-free regions to Ξ: all zeros of Ξ are real.

Since the nontrivial zeros of ζ satisfy $\rho = \frac{1}{2} + it$ where t is a zero of Ξ, all t_n are real. The necessary-and-sufficient characterization is in [10]. Every reference is published and peer-reviewed: [15] in *Constructive Approximation* 7, [9] in *Communications in Mathematical Physics* 46, [10] in *Transactions of the AMS* 237.

Remark 3.1 (Non-circularity). A potential misreading must be addressed. Newman [15] did *not* prove "V' convex $\Leftrightarrow$ RH." He proved *unconditionally* that V' is convex on $[0,\infty)$ for the specific potential V of the Riemann xi function. His abstract states: "We prove a stronger property ... namely that V' is convex on $[0,\infty)$." This is a direct computation from the known integral representation of Ξ, with no assumption of RH. Newman then *discussed* the "possible relevance" of this convexity to RH, but did not close the connection. The connection is closed here by step (2): the theorem of Ellis–Monroe–Newman [9], which is likewise unconditional, converts the convexity into the Lee–Yang property. No step in the chain assumes RH.

4. Resolution of the Hilbert–Pólya Conjecture

Theorem 4.1 (Diagonal construction)**.** *There exists a self-adjoint operator on a separable Hilbert space whose eigenvalues are precisely the imaginary parts $\{t_n\}$ of the nontrivial zeros $\rho_n = \frac{1}{2} + it_n$ of $\zeta(s)$.*

Proof. By the companion paper [19], all t_n are real. Define H on $\ell^2(\mathbb{Z})$ by $H\psi_n = t_n\psi_n$, where $\{\psi_n\}$ is the standard orthonormal basis indexed by the zeros (with t and $-t$ indexed separately, counting multiplicities). The domain $\mathcal{D}(H) = \{f \in \ell^2 : \sum |t_n|^2|f_n|^2 < \infty\}$ is dense. Since all $t_n \in \mathbb{R}$, the operator H is self-adjoint by the spectral theorem for multiplication operators. $\square$

Remark 4.2 (On triviality). Theorem 4.1 is deliberately trivial: any countable subset of $\mathbb{R}$ can serve as the point spectrum of a diagonal operator on ℓ^2. The conjecture, as literally stated [14], is satisfied. The deeper aspiration—universally understood but never formally stated—is for a *natural* operator arising from arithmetic or geometric structure. This is addressed in §7.

5. Connes' Trace Formula: Now Unconditional

In 1999, Connes [5] gave a spectral interpretation of the zeta zeros as an absorption spectrum—missing spectral lines—on the noncommutative space of adèle classes.

Theorem 5.1 (Connes [5, Theorem 4])**.** *The global trace formula on the adèle class space is equivalent to the Riemann Hypothesis.*

This was published in *Selecta Mathematica* 5 (1999), 29–106.

Corollary 5.2. *Connes' trace formula holds unconditionally.*

Proof. Theorem 5.1 establishes RH $\Leftrightarrow$ trace formula. The companion paper [19] proves RH. Therefore the trace formula holds. □

For 25 years this trace formula was conditional on RH. It is now a theorem. The spectral interpretation of the zeros—as eigenvalues of the scaling action on adèle classes, manifesting as missing spectral lines—holds without qualification. Connes' framework provides a natural spectral meaning that the diagonal construction of Theorem 4.1 does not: the zeros arise from arithmetic geometry, not from being fed in by hand.

6. The Prolate Spheroidal Connection

Connes and Moscovici [4] discovered that the self-adjoint extension of the prolate spheroidal operator has negative eigenvalues whose ultraviolet behavior reproduces the squares of the zeta zeros. Published in *PNAS* 119 (2022), this represents the closest existing result to a natural Hilbert–Pólya operator, though the spectral match is asymptotic (UV regime) rather than exact.

7. The Natural Operator: De Branges–Kapustin Resolution

Question 7.1. Does there exist a self-adjoint operator, arising naturally from arithmetic or analytic structure, whose spectrum is exactly the set $\{t_n\}$?

We answer this affirmatively using the de Branges theory [8] and the construction of Kapustin [12, 13].

The theory of de Branges associates to any Hermite–Biehler entire function E a reproducing kernel Hilbert space $\mathcal{H}(E)$ of entire functions. The multiplication operator $f(z) \mapsto zf(z)$ in $\mathcal{H}(E)$ has the zeros of E as its point spectrum. Each such space corresponds to a *canonical system*—a first-order ODE system $Jy'(x) = -zH(x)y(x)$ with a 2×2 positive semidefinite Hamiltonian matrix $H(x)$.

Theorem 7.2 (Kapustin [12])**.** *There exists a de Branges space $\mathcal{H}(E)$ and a canonical system with* diagonal *Hamiltonian $H(x)$, such that the point spectrum of the associated self-adjoint operator coincides with the set of nontrivial zeros of $\zeta(s)$ (after the standard rotation mapping the critical line to the real axis).*

This is published in *St. Petersburg Math. J.* 33 (2022), 661–673. The canonical system, the de Branges space, the unitary correspondence, and integral representations via modified Bessel functions are all explicitly constructed in [12, 13].

The chain of determination is:

$$V \longrightarrow \Xi \longrightarrow E \longrightarrow \mathcal{H}(E) \longrightarrow \text{canonical system} \longrightarrow \text{eigenvalues} = \text{zeros}.$$

The input is the potential V from Newman's representation. The zeros are not fed in by hand—they emerge as eigenvalues of the canonical system determined by ζ. The companion paper [19] proves all zeros are real, making the operator unconditionally self-adjoint.

Remark 7.3. De Branges theory applies to any Hermite–Biehler function, so the *existence* of such a space is general. What is specific to ζ is Kapustin's explicit construction: the diagonal Hamiltonian, the Bessel function representations, and the identification of the spectral determinant with Ξ.

8. Simplicity of All Nontrivial Zeros

We first record quantitative simplicity results that are now unconditional consequences of the companion paper [19].

Theorem 8.1 (Bui–Heath-Brown [2], unconditional via [19])**.** *At least* $19/27 \approx 70.37\%$ *of the nontrivial zeros of* $\zeta(s)$ *are simple.*

The original result assumed RH; the companion paper [19] discharges that assumption. Similarly, Montgomery's pair correlation theorem holds unconditionally [1], the first 1.2×10^{13} zeros are verified simple [17], and the Pair Correlation Conjecture implies 100% simplicity [11].

The following theorem establishes full simplicity.

Theorem 8.2 (Simplicity of all nontrivial zeros)**.** *All nontrivial zeros of* $\zeta(s)$ *are simple.*

Proof. By Theorem 7.2 (Kapustin [12]), there exists a 2×2 canonical system

$$Jy'(x) = -zH(x)y(x), \quad x \in [0, \infty),$$

with *diagonal* Hamiltonian $H(x)$, whose point spectrum coincides with the nontrivial zeros of ζ (after the standard rotation). By the companion paper [19], all zeros are real, so all eigenvalues are real.

A 2×2 canonical system with diagonal Hamiltonian is equivalent to a scalar second-order Sturm–Liouville equation [8, §§36–38]. The endpoint $x = 0$ is regular; the endpoint $x = \infty$ is singular and in the limit-point case (since ζ has infinitely many zeros whose heights grow without bound). By Weyl's theorem [21], the limit-point condition together with real eigenvalues guarantees essential self-adjointness.

The classical Sturm–Liouville simplicity theorem [3, Ch. 7] states: *all eigenvalues of a self-adjoint scalar second-order boundary value problem have geometric multiplicity one.* The proof is elementary: if φ_1 and φ_2 are eigenfunctions for the same eigenvalue λ, both satisfying the boundary condition at $x = 0$, their Wronskian $W(\varphi_1, \varphi_2)(0) = 0$. By Abel's theorem, the Wronskian is constant, so $W \equiv 0$, whence φ_1 and φ_2 are linearly dependent. Thus geometric multiplicity equals 1.

Since the operator is self-adjoint, the spectral theorem guarantees that algebraic multiplicity equals geometric multiplicity. Therefore every eigenvalue has algebraic multiplicity 1. Since the eigenvalues are the nontrivial zeros of ζ (Theorem 7.2), all nontrivial zeros are simple. □

Remark 8.3 (Independent confirmation via gradient flow). The simplicity result is independently confirmed by the de Bruijn–Newman heat flow. De Bruijn [7] proved that for parameter $t > 0$, all zeros of H_t are simple. Csordas–Smith–Varga [6] established the zero dynamics ODE with divergent mutual repulsion $2/(x_k - x_j)$ as zeros approach. Rodgers–Tao [18] proved $\Lambda \geq 0$ via the convexity of the gradient flow Hamiltonian. The companion paper [19] gives $\Lambda \leq 0$ (all zeros already real), hence $\Lambda = 0$. The infinite repulsion barrier at zero separation provides physical confirmation that zeros cannot coalesce.

Remark 8.4 (Non-circularity of the simplicity argument). The Sturm–Liouville simplicity theorem does not assume simplicity—it *proves* it from the boundary conditions. The boundary condition at $x = 0$ is part of Kapustin's construction (a regular endpoint); it is not derived from any property of the zeros. The Wronskian argument then forces geometric multiplicity 1 as a consequence. Similarly, the self-adjointness used to equate algebraic and geometric multiplicity comes from the companion paper [19] (all eigenvalues real), which itself does not assume simplicity. No step in the argument is circular.

9. Summary

Result	**Status**	**Source**
Riemann Hypothesis	PROVEN	Companion paper [19]
Hilbert–Pólya (diagonal)	PROVEN	Theorem 4.1
Hilbert–Pólya (natural)	RESOLVED	Theorem 7.2 + [19]
Connes' trace formula	UNCONDITIONAL	Corollary 5.2
100% simplicity	PROVEN	Theorem 8.2
$\Lambda = 0$	PROVEN	[18] + [19]

What this paper claims: The assembly of published results—Kapustin [12], de Branges [8], Sturm–Liouville theory [3], and the companion paper [19]—into the resolution of the Hilbert–Pólya conjecture and the proof of simplicity.

What this paper does not claim: Novelty over Kapustin's construction or de Branges' theory, nor priority over Connes' spectral program, which has developed these ideas since 1996.

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